

Behind the scenes of The Cavern UE5 Cinematic Visual Tech Test

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The Coalition presents a behind the scenes look at ‘The Cavern’, a Unreal Engine 5 Cinematic Visual Tech Test that was created for internal validation purposes and shown publicly at Unreal Engine 5’s Release Stream ‘The State of Unreal 2022 Keynote’. This cinematic was created to validate our internal next-gen modeling process, our animation process, as well as gather performance metrics on Xbox Series X.

For modeling the character was modeled at 4x the resolution of an Xbox One character, landing around 200,000 triangles. The real difference in workflow was the replacement of hair ‘cards’ with UE5’s groom assets. The groom assets were generated using X-Gen for Maya and exported as Alembic files. The groom assets for the head included hair, beard, eyebrows, eyelashes, peach fuzz and totaled about 3.5 million triangles.

The head was modeled in Maya and Z-brush. Eyes, mouth-bag, and teeth were generated using MetaHumans and exported as fbx files into the Maya scene. Skin material was based on the MetaHumans skin shader leveraging Burley SSSSS.

The custom face rig was driven by UE5’s ControlRig. The rig was comprised of morph targets, PoseAsset driven Bone transforms, and shader parameters (for wrinkles). The control rig was generated using python scripts to allow for quick iteration. The face rig consisted of 12 metahuman based wrinkle regions, and 3 normal map wrinkle regions. We had 126 blendshapes on the head, 81 for the beard, 38 for eyelashes, eye AO geo, and eye water geo respectively. Blendshapes were generated using 4d head-scan data of a different actor and re-targeted to the custom modeled face, which isn’t recommended but was what worked for us logistically given the timelines. The body consisted of 60 animated body joints, 54 procedurally driven secondary rigging body joints, and 44 face joints used for offsetting blendshapes.

For the animation process we broke it down into a few specific capture steps. First the actor was captured doing their performance at Side LA, where we captured the facial performance and audio. In

our studio we had the animation team act out the sequence to their liking wearing a motion capture suit and captured the performance in real-time in engine leveraging UE5’s Virtual Production workflow. The body motion and facial performance, were combined by Cubic Motion by tracking the performance onto the rig The Coalition supplied. We implemented a bi-directional export pipeline that allows us to export anim data from Unreal back into Motion Builder and then export back into Unreal losslessly to allow us to iterate where the animators were most comfortable.

The packs on the waist and the collar used our proprietary Rod Dynamics simulation which simulates bouncing of joints on the CPU. The flap of cloth over the pelvis on the character was simulated using the UE5 Chaos Cloth System.

In terms of performance the demo runs at 25 to 40fps on Xbox Series X. The primary cost is the groom assets and shadowing of the groom assets taking around 9 milliseconds. We have since optimized our groom asset workflow process to reduce strand count which reduces GPU cost by 33% and simplifying our hair shader. Secondary cost is Epic quality Lumen at 7ms. Virtual Shadows are 5.5ms. The cost is higher for Virtual Shadows on average for cinematics as they are affecting a large-on-screen non-nanite asset (a skeletal mesh), so they lose all built-in acceleration that comes from virtual shadows and Nanite.

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